

MEARN Assignment

(1) What is web development?

→ Web development is the work of creating, designing, and maintaining websites like google, amazon, facebook, or any normal website you use daily.

It includes:

1. front-end development
2. Back-end development
3. database

(2) Difference between frontend and backend with examples.

Frontend

• frontend is the visible part of a website.

- what users see and use
- Runs in the browser
- focuses on design & user experience

Back-end

• Backend is the hidden part of a website.

- works behind the scenes
- Runs on the server
- focuses on logic & data handling.

example:

Frontend = user interface (UI)

- Buttons
- Text

example:

Backend = ^{server} ~~less~~ -side work

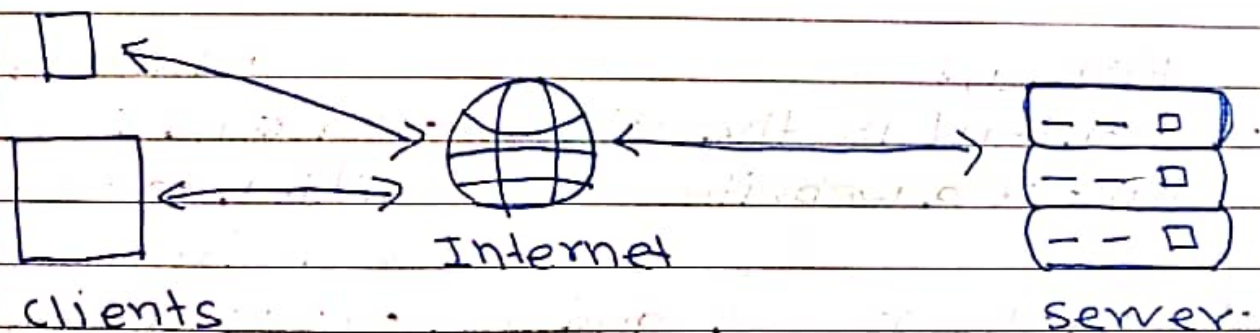
- saving user data
- fetching data from db

(3) explain client-server communication with a neat diagram.

→ client-server communication is a process where:

- Client sends a request.
 - Server receives the request, processes it, and
 - Server sends back a response.
- A client (browser or app) asks for data, and the server gives the data back.

Diagram:



b) What is MEAN Stack?

→ MEAN Stack is a full-stack web development technology stack used to build modern web applications.

MEAN = MongoDB, Express, Angular, Node.js

MEAN Stack is popular for:

- Uses JavaScript everywhere
- Fast performance

• Good for single-page applications (SPA).

Install Angular CLI using npm install -g @angular/cli

To install Angular CLI, use this command:
npm install -g @angular/cli

Step-by-Step explanation:

Step 1: install Node.JS

Angular CLI needs Node.JS

Node.JS already includes npm

check installation:

node -v

npm -v

Step 2: install Angular CLI

Run this command in command prompt/terminal

npm install -g @angular/cli

npm: Node package manager

install: installs the package

-g → installs globally (use anywhere)

@angular/cli: Angular command line interface

Angular CLI is used for:

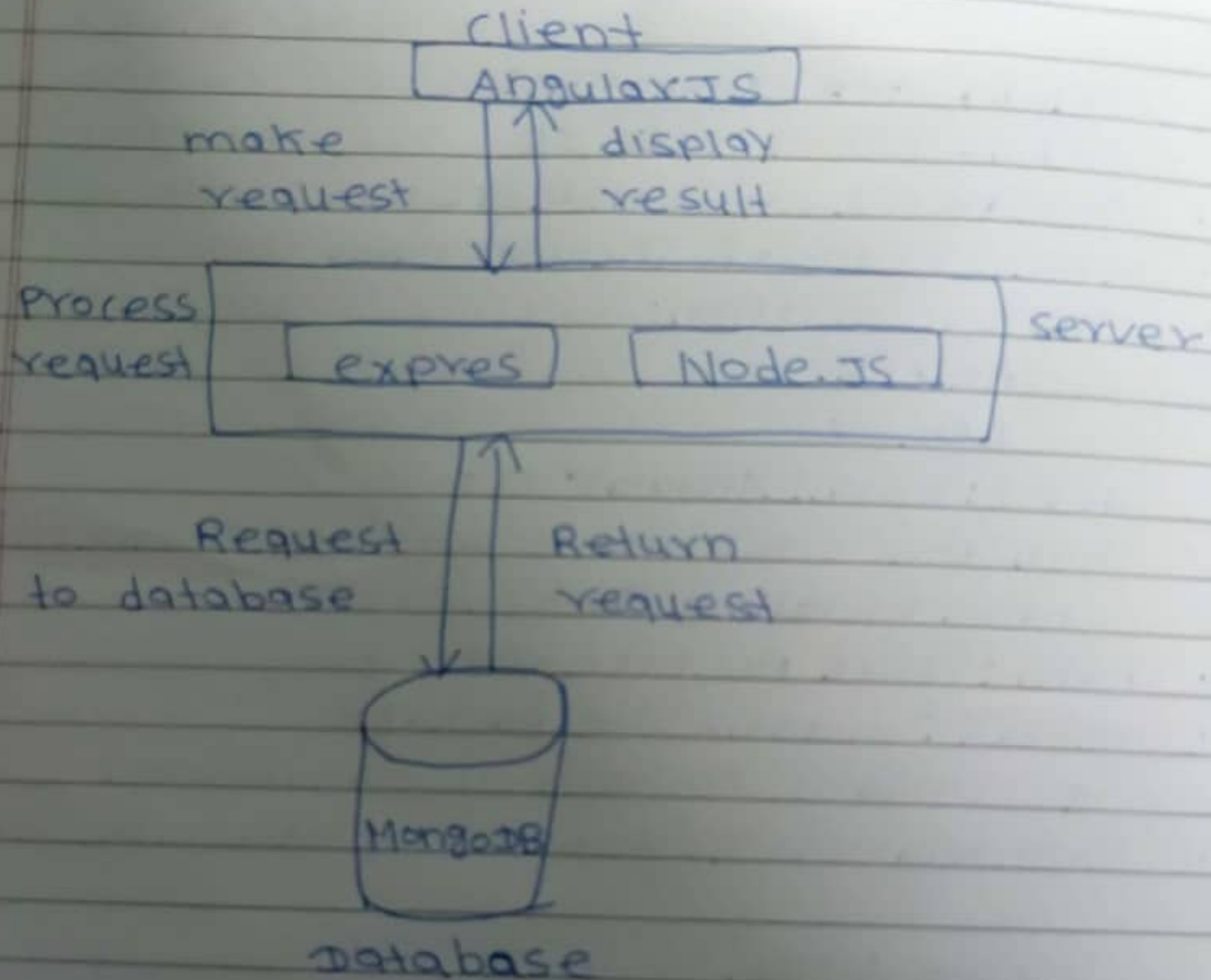
Creates Angular projects easily

Runs and builds applications

Saves time and effort.

(6) Draw the MEAN architecture workflow:
client (Angular) → server (express + Node.js) → Database (MongoDB).

→ workflow:



1. client (Angular):
 - user clicks button / submits form
2. server (Node.js + Express)
 - Receives request
 - processes logic
3. database (MongoDB)
 - stores and retrieves data
4. Response flow
 - MongoDB
 - data sent

Saved successfully